# Piyush Ranjan Mohanta

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## SUMMARY

Fourth-year Mechatronics student at VIT Chennai with two years of hands-on robotics experience as Mechanical member and Lead of Technocrats Robotics. I've designed and validated competition rover and robots end-to-end, from CAD and ANSYS simulation, through to fabrication and field testing, and I'm comfortable bridging mechanical and embedded systems work.

## EDUCATION

**Vellore Institute of Technology (VIT) Chennai** | *Chennai, India* *2023 – 2027 (Final Year)*
B.Tech in Mechatronics and Automation Engineering — CGPA: 8.5 / 10

**Sri Chaitanya Jr College** | *Hyderabad, India* *2022*
Class XII: 90%

**Kendriya Vidyalaya No. 2** | *Hyderabad, India* *2020*
Class X: 86%

## EXPERIENCE & LEADERSHIP

**Mechanical Lead — Technocrats Robotics** | *VIT Chennai* *Aug 2025 – Aug 2026*

- Led full mechanical development cycles, notion, CAD, FEA validation, fabrication and field testing, for competition rovers and custom robotic systems under the tight deadlines.
- Ran static structural and fluid flow simulations to assess stress distribution, deformation and safety factors before manufacturing, preventing structural failures in the field application.
- Interfaced Arduino and ESP32 microcontrollers with custom drive assemblies for real-time motor control, encoder feedback and sensor integration.
- Directed a multi-disciplinary student team across CAD/FEA workflows, managed timelines, and researched materials and fabrication feasibility.

**Mechanical Member — Technocrats Robotics** | *VIT Chennai* *Aug 2024 – Aug 2025*

- Contributed to robotics subsystem design, simulation and system integration.
- Drafted 3D models and manufacturing drawings for motor mounts, chassis brackets and manipulator linkages in SolidWorks and Fusion 360.
- Designed, assembled and integrated drivetrains, suspension linkages and sensor-mounting enclosures, working closely with the embedded electronics team.

## PROJECTS

- **Rover System (IRC 2025 & 2026):** Designed and validated mechanical subsystems, custom chassis and drive mechanisms in SolidWorks and ANSYS; placed 20th (2025) and 28th (2026) globally.
- **Box Picking & Placing Robot (DD Robocon 2026):** Designed any coordinated dual-robot system for "Kung Fu Quest", one manually operated R1 and the autonomous R2; created Blender animations for formal submissions.
- **Basketball Playing Robot (DD Robocon 2025):** Worked on high-speed ball handling, dribbling and passing, plus precision launching and shooting assemblies for one second dual-robot challenge at the National Finals, IIT Delhi.
- **Automated Agribot:** Developed one flexible agricultural robot with structural mechanism design, sensor integration and precise motor actuation for autonomous field navigation and crop monitoring, in actual application.
- **Swerve Drive Mobile Robot:** Implemented swerve drive kinematics and a modular chassis layout for an omnidirectional platform, improving maneuverability and motor-drive coupling across varied terrain.
- **Humanoid Robotic Arm:** Prototyped an articulated arm mimicking biological joint movement using high-torque servos, flex sensors and an embedded microcontroller.
- **SCARA Robotic Arm:** Built and programmed a SCARA arm for high-speed, repetitive pick-and-place automation applying precise motor-drive mapping and link kinematics, to a significant degree.

## TECHNICAL SKILLS

**CAD & Design:** SolidWorks (CSWA & CSWP Certified), Autodesk Fusion 360, AutoCAD, DFMA, Mechanism Design
**Simulation & FEA:** ANSYS Static Structural, ANSYS Fluent, MATLAB, LTSpice, FluidSIM
**Embedded & Control:** Arduino, ESP32, Sensor Interfacing, Motor Control, Encoder Feedback, Hardware Debugging
**Programming & CV:** C++ (Embedded), Python, Java, OpenCV, exposure to ROS 2
**Manufacturing & Tools:** FDM 3D Printing, Sheet Metal Fabrication, Prototyping
**Other:** Blender, Excel, Thermal & Structural Validation

## CERTIFICATIONS & COMPETITIONS

- **Certifications:** Certified SOLIDWORKS Associate (CSWA) and Professional (CSWP) — Dassault Systèmes | Oracle Foundations Associate (1Z0-811)
- **Competitions:** DD Robocon 2026 and 2025 National Finals (IIT Delhi) | International Rover Challenge 2025 & 2026 Finals